



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	PHYSICAL CHEMISTRY IV – STATISTICAL THERMODYNAMICS				
Type of Course	MAJOR				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	CHE2CJ102, CHE3CJ201 and CHE4CJ205. Preliminary idea about quantum chemistry, classical thermodynamics and kinetics				
Course Summary	The bulk properties of matter are linked with its microscopic properties and it is important to consider a molecular approach to quantitatively explain the physical and chemical properties. This course explores the basics of statistical thermodynamics and thermodynamics of irreversible processes and solutions, molecular dynamics theories.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To Understand and apply concepts of statistical thermodynamics and quantum statistics in chemical applications	U	C	Test /Seminar/Assignment
CO2	To Calculate thermodynamic properties using partition functions	U	P	Test/ Assignment/Seminar
CO3	To Apply thermodynamic principles to solutions	An	P	Quiz/Seminar/Assignment

	and comprehend irreversible processes			
CO4	To understand theories of molecular reaction dynamics	U	C	Test/Seminar /Assignment
CO5	To analyse potential energy surfaces in order to understand reaction dynamics	An	C	Assignment/Test
CO6	To apply concepts of statistical thermodynamics in simulation experiments	Ap	M	Lab work/Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	STATISTICAL THERMODYNAMICS- I		15	26
	1	Fundamentals: Concept of distribution, thermodynamic probability and most probable distribution.	2	
	2	Ensembles, statistical mechanics for systems of independent particles and its importance in chemistry	2	
	3	Thermodynamic probability & entropy, idea of microstates and macrostates, statistical weight factor (g)	2	
	4	Sterling approximation, and Maxwell- Boltzmann distribution of molecular energies.	2	
	5	The molecular partition function and its relation to the thermodynamic properties, derivation of third law of thermodynamics, equilibrium-constant & equi-partition principle in terms of partition functions, factorisation of the molecular partition function into translational, rotational, vibrational and electronic parts, the corresponding contributions to the thermodynamic properties.	4	
	6	Relation between molecular & molar partition functions, Evaluation of partition functions and thermodynamic properties for ideal mono-atomic and diatomic gases.	3	
II	STATISTICAL THERMODYNAMICS- II		10	26
	7	Quantum Statistics: Bose-Einstein distribution law, Bose-Einstein Condensation, application to liquid helium.	3	

	8	Fermi - Dirac distribution law, application to electrons in metals.	2	
	9	Relationship between Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac statistics.	2	
	10	Heat capacities of solids: Classical and quantum theories, Einstein's theory of atomic crystals and Debye's modification.	3	
III	THERMODYNAMICS OF SOLUTIONS AND IRREVERSIBLE PROCESSES		15	30
	11	Fugacity, Activity, Activity coefficient, Standard state of substance (for solute and solvents), Duhem-Margules equation and its applications	3	
	12	Thermodynamics of ideal solutions, Deduction of the laws of Raoult's ebullioscopy, cryoscopy, and osmotic pressure.	2	
	13	Non ideal solutions, Deviations from Raoult's law.	2	
	14	Excess functions: Excess free energy, excess entropy, excess enthalpy, excess volume.	2	
	15	Simple examples of irreversible processes, general theory of non-equilibrium processes	2	
	16	Entropy production, The phenomenological relations, Onsager reciprocal relations	2	
	17	Application to the theory of diffusion, thermal diffusion, thermo-osmosis and thermo- molecular pressure difference, electro-kinetic effects, the Glansdorf-Pregogine equation.	2	
IV	MOLECULAR REACTION DYNAMICS		5	16
	18	Reactive encounters: Collision theory	1	
	19	Diffusion controlled reactions	1	
	20	The material balance equation	1	
	21	Potential energy surfaces: Attractive and repulsive surfaces	1	
	22	Theories of unimolecular reactions: Rice -Ramsperger and Kassel (RRK) model.	1	
V	Practicals		30	
	A minimum of three experiments/simulations must be performed and recorded			
			5	

	1. By using MS Excel, Scilab, Python or any other suitable programs, simulate and plot the probability of macrostates in coin tossing experiments 2. By using a suitable computer program (MS Excel, Scilab or other) determine and plot Maxwell speed distribution function for different gases 3. By using a suitable computer program (MS Excel, Scilab or other), determine and plot Maxwell-Boltzmann, Bose-Einstein & Fermi-Dirac distribution functions 4. Simulation of specific heat of solids (diamond) using Dulong Petit, Einstein and Debye model 5. Scan around a single bond in N₂ molecule to find out the minimum energy in the PE diagram by using Gaussian, ORCA, Gamess or similar packages. (HF method, 6-31G basis set) 6. Calculation of Potential energy surface of H₂O molecule with respect to change of bond angles and bond distances by using Gaussian, Gamess or similar packages (HF method, 6-31G basis set) 7. Calculation of Potential Energy Surface of ethane by changing the dihedral angle, in order to understand the most stable conformation by using Gaussian, Gamess or similar packages 8. Calculation of Potential Energy Surface of n-butane by changing the dihedral angle, in order to understand the most stable conformation by using Gaussian, Gamess or similar packages 9. Open ended 10. Open ended	5 5 5 5 5 5 5 5 5 5	
	References Module I and II 1. Physical Chemistry: Thermodynamics, Structure and Change, 10th Edition, P. Atkins and J. de Paula, (W. H Freeman and Company, New York) 2. D. A. McQuarrie, J. D. Simon, Physical Chemistry – A Molecular Approach, (Viva, 2001.) 3. T. Engel, P. Reid, Thermodynamics, Statistical Thermodynamics & Kinetics, Pearson Education, Inc: New Delhi, 2007. 4. J. Rajaram, J. C. Kuriacose, Chemical Thermodynamics, Classical, Statistical and irreversible, Pearson Education, New Delhi, 2013. 5. A. Ben-Naim, Statistical Thermodynamics Based on Information: A Farewell to Entropy, World Scientific, Singapore		

		Module III and IV - Physical Chemistry: Thermodynamics, Structure and Change, 10th Edition, P. Atkins and J. de Paula, (W. H Freeman and Company, New York) D. A. McQuarrie, J. D. Simon, Physical Chemistry – A Molecular Approach, (Viva, 2001.) T. Engel, P. Reid, Thermodynamics, Statistical Thermodynamics & Kinetics, Pearson Education, Inc: New Delhi, 2007. J. Rajaram, J. C. Kuriacose, Chemical Thermodynamics, Classical, Statistical and irreversible, Pearson Education, New Delhi, 2013. K. Laidler, Chemical Kinetics, 3rd Edn., Pearson Education, New Delhi, 2004. Module V https://www.scilab.org/ Wolfram Demonstrations Project https://www.orcasoftware.de/tutorials_orca/# - Advanced Physical Chemistry: Practical Guide, C. Arora and S. Bhattacharya, Bentham Books, UAE, 2022 H. Singh, <i>Resonance</i>, December 1996, Page 49-59, A Simple Experiment to Study the Statistical Properties of a Molecular Assembly with Two or Three State Dynamics Further reading - G.S. Rush Brooke, <i>Statistical mechanics</i>, Oxford University Press. - T.L. Hill, <i>Introduction to statistical thermodynamics</i>, Addison Wesley. K. Huary, <i>Statistical mechanics, Thermodynamics and Kinetics</i>, John Wiley. - O.K. Rice, <i>Statistical mechanics, Thermodynamics and Kinetics</i>, Freeman and Co. - F.C. Andrews, <i>Equilibrium statistical mechanics</i>, John Wiley and sons, 1963. M. C. Guptha, <i>Statistical Thermodynamics</i>, Wiley eastern Ltd., 1993 - Pigoggine, <i>An introduction to Thermodynamics of irreversible processes</i>, Interscience Publisher - B.G. Kyle, <i>Chemical and Process Thermodynamics</i>, 2nd Edn, Prentice Hall of India - K.J. Laidler, J.H. Meiser and B. C. Sanctuary, <i>Physical Chemistry</i>, Houghton Mifflin Company, New York, 2003. - Richard I. Masel, <i>Chemical Kinetics and Catalysis</i>, Wiley Interscience, 2001	
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Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O 5	PS O 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	-	-	3	2	3	2	2	-	2	-	1
CO 2	3	2	-	-	3	2	3	2	1	-	2	-	1
CO 3	3	2	-	-	3	2	3	2	1	-	2	-	1
CO 4	3	2	1	-	3	3	3	2	1	-	1	-	1
CO 5	3	3	2	1	3	3	3	2	1	-	3	-	1
CO 6	3	3	1	3	3	3	3	2	1	2	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment /viva	Practical skill evaluation	End Semester Examinations
CO 1		✓		✓
CO 2		✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓	✓	✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓